

2.12 Application of Fourier Transforms: Filters

In a variety of applications, it is of interest to change the relative amplitudes of the frequency components in a signal or perhaps eliminate some frequency components entirely, a process referred to as *filtering*. Linear time-invariant systems that change the shape of the spectrum are often referred to as *frequency-shaping filters*. Systems that are designed to pass some frequencies essentially undistorted and significantly attenuate or eliminate others are referred to as *frequency-selective filters*. Consequently, filtering can be conveniently accomplished through the use of LTI systems with an appropriately chosen frequency response, and frequency-domain methods provide us with the ideal tools to examine this very important class of applications.

Frequency-Selective Filters

Frequency-selective filters are a class of filters specifically intended to accurately or approximately select some bands of frequencies and reject others. The use of frequency-selective filters arises in a variety of situations. For example, if noise in an audio recording is in a higher frequency band than the music or voice on the recording is, it can be removed by frequency-selective filtering. Another important application of frequency-selective filters is in communication systems. As we discuss in detail later, the basis for amplitude modulation (AM) systems is the transmission of information from many different sources simultaneously by putting the information from each channel into a separate frequency band and extracting the individual channels or bands at the receiver using frequency-selective filters. Frequency-selective filters for separating the individual channels and frequency-shaping filters for adjusting the quality of the tone form a major part of any home radio and television receiver.

While frequency selectivity is not the only issue of concern in applications, its broad importance has led to a widely accepted set of terms describing the characteristics of frequency-selective filters. In particular, while the nature of the frequencies to be passed by a frequency-selective filter varies considerably from application to application, several basic types of filter are widely used and have been given names indicative of their function. For example, a *lowpass filter* is a filter that passes low frequencies—i.e., frequencies around $\omega = 0$ —and attenuates or rejects higher frequencies. A *highpass filter* is a filter that passes high frequencies and attenuates or rejects low ones, and a *bandpass filter* is a filter that passes a band of frequencies and attenuates frequencies both higher and lower than those in the band that is passed. In each case, the *cutoff frequencies* are the frequencies defining the boundaries between frequencies that are passed and frequencies that are rejected—i.e., the frequencies in the *passband* and *stopband*.

Numerous questions arise in defining and assessing the quality of a frequency-selective filter. How effective is the filter at passing frequencies in the passband? How effective is it at attenuating frequencies in the stopband? How sharp is the transition near the cutoff frequency—i.e., from nearly free of distortion in the passband to highly attenuated in the stopband? Each of these questions involves a comparison of the characteristics of an actual frequency-selective filter with those of a filter with idealized behavior. Specifically, an *ideal frequency-selective filter* is a filter that exactly passes complex exponentials at one set of frequencies without any distortion and completely rejects signals at all other frequencies. For example, a continuous-time *ideal low pass filter* with cutoff frequency ω_c is an LTI system that passes complex exponentials $e^{j\omega t}$ for values of ω in the range $-\omega_c \leq \omega \leq \omega_c$ and rejects signals at all other frequencies. That is, the frequency response of a continuous-time ideal lowpass filter is

$$H(\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & |\omega| > \omega_c \end{cases}, \quad (2.40)$$

as shown in **Figure 2.41**.

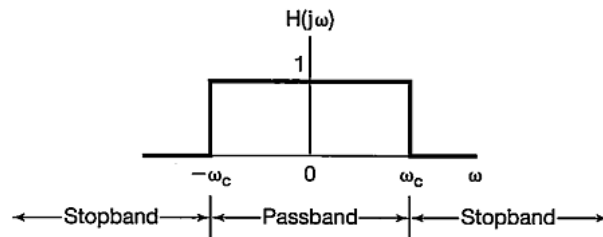


Figure 2.41. Frequency response of an ideal low-pass filter.

Figure 2.42 (a) depicts the frequency response of an ideal continuous-time high-pass filter with cutoff frequency ω_c , and **Figure 2.42 (b)** illustrates an ideal continuous-time band-pass filter with lower cutoff frequency ω_{c1} and upper cutoff frequency ω_{c2} . Note that each of these filters is symmetric about $\omega = 0$, and thus, there appear to be two passbands for the high-pass and band-pass filters. This is a consequence of our having adopted the use of the

complex exponential signal $e^{j\omega t}$, rather than the sinusoidal signals $\sin\omega t$ and $\cos\omega t$, at frequency ω . Since $e^{j\omega t} = \cos\omega t + j \sin\omega t$, both of these complex exponentials are composed of sinusoidal signals at the same frequency ω . For this reason, we usually define ideal filters so that they have the symmetric frequency response behavior seen in **Figure 2.41** and **Figure 2.42**.

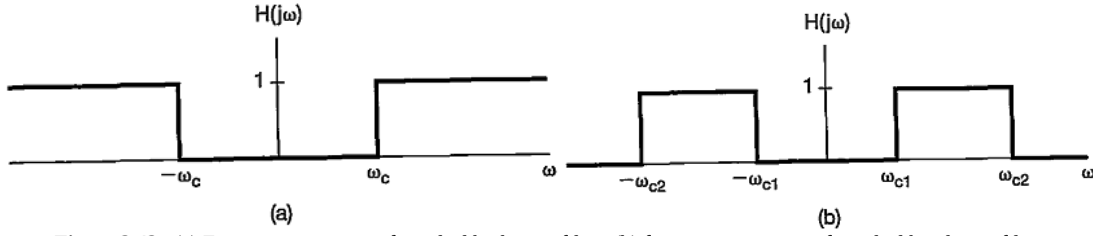


Figure 2.42. (a) Frequency response of an ideal high-pass filter; (b) frequency response of an ideal band-pass filter.

As we will see on numerous occasions, ideal filters are quite useful in describing idealized system configurations for a variety of applications. However, they are not realizable in practice and must be approximated. Furthermore, even if they could be realized, some of the characteristics of ideal filters might make them undesirable for particular applications, and a non-ideal filter might in fact be preferable.

In detail, the topic of filtering encompasses many issues, including design and implementation. While we will not delve deeply into the details of filter design methodologies, in the remainder of this chapter and the following chapters we will see a number of other examples of both continuous-time and discrete-time filters and will develop the concepts and techniques that form the basis of this very important engineering discipline.

2.12.1. Example of a real practical low-pass filter

Example 2.23

For the circuit shown below, determine (a) the transfer function; (b) the impulse response.

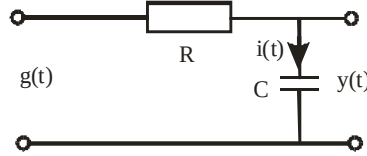


Figure 2.43. A practical low-pass filter. It can also be viewed as a butterworth filter with order $N=1$.

Solution:

From circuit theory, we have
$$g(t) = i(t)R + y(t) \quad \text{and} \quad i(t) = C \frac{dy(t)}{dt}.$$

Thus,
$$RC \frac{dy(t)}{dt} + y(t) = g(t)$$

Taking the Fourier transform, we have
$$j\omega RC Y(\omega) + Y(\omega) = G(\omega)$$

(a) The transfer function is:
$$H(\omega) = \frac{Y(\omega)}{G(\omega)} = \frac{1}{1 + j\omega RC}$$

(b) From the Fourier transform pair,
$$e^{-at}u(t) \Leftrightarrow \frac{1}{a + j\omega}$$

we have,
$$h(t) = \begin{cases} \frac{1}{RC} e^{-\frac{t}{RC}} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

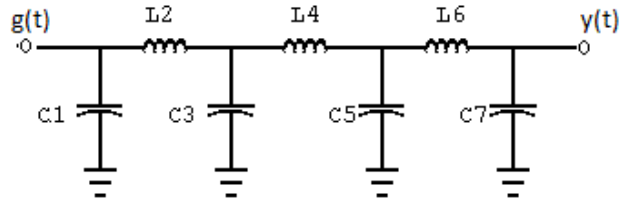


Figure 2.44. A butterworth low-pass filter with order $N=7$.

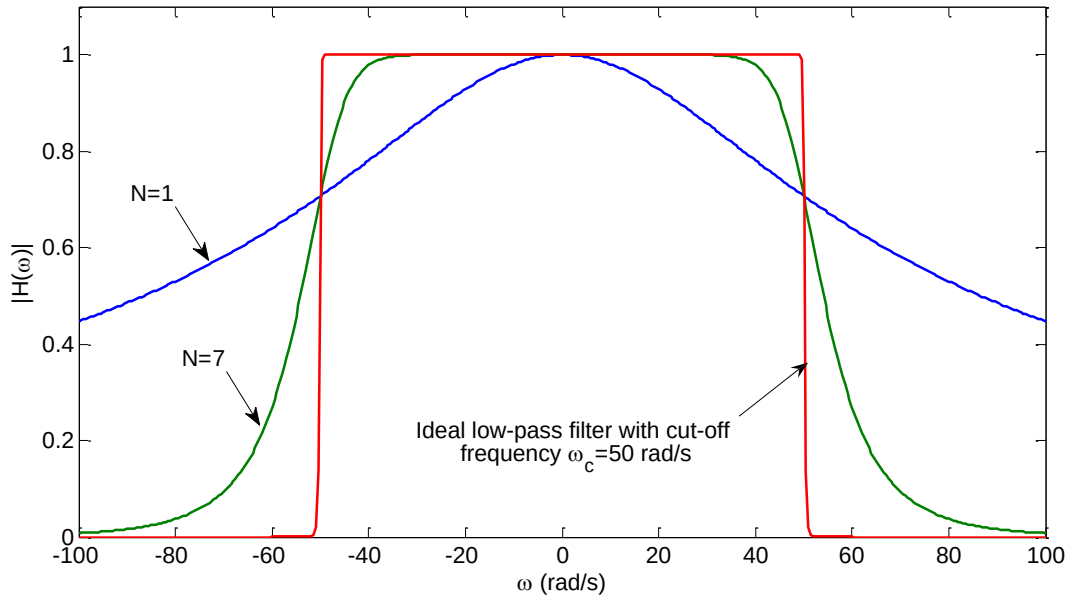


Figure 2.45. Transfer function characteristics for butterworth low-pass filters with different orders. The cutoff frequency is 50 rad/s.

2.12.2. Frequency-Selective Filtering with Variable Center Frequency

For example, consider the system shown in **Figure 2.46**. Here, an input signal $x(t)$ is multiplied by the complex exponential signal $e^{j\omega_c t}$. The resulting signal is then passed through a lowpass filter with cutoff frequency ω_0 , and the output is multiplied by $e^{-j\omega_c t}$. The spectra of the signals $x(t)$, $y(t)$, $w(t)$, and $f(t)$ are illustrated in **Figure 2.47**.

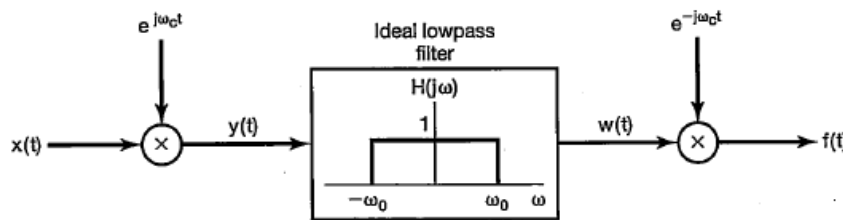


Figure 2.46. Implementation of a band-pass filter using amplitude modulation with a complex exponential carrier.

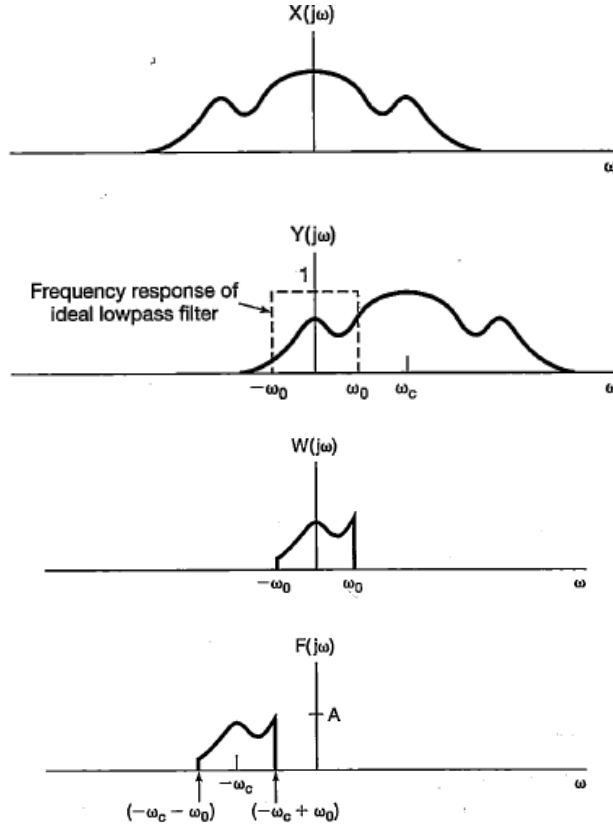


Figure 2.47. Spectra of the signals in the system of **Figure 2.46**.

Specifically, from either the multiplication property or the frequency-shifting property it follows that the Fourier transform of $y(t) = e^{j\omega_c t} x(t)$ is

$$Y(\omega) = \int_{-\infty}^{+\infty} \delta(\theta - \omega_c) X(\omega - \theta) d\theta$$

So that $Y(\omega)$ equals $X(\omega)$ shifted to the right by ω_c and frequencies in $X(\omega)$ near $\omega = \omega_c$ have been shifted into the passband of the low-pass filter. Similarly, the Fourier transform of $f(t) = e^{-j\omega_c t} w(t)$ is

$$F(\omega) = W(\omega + \omega_c),$$

so that the Fourier transform of $F(\omega)$ is $W(\omega)$ shifted to the left by ω_c . From **Figure 2.47**, we observe that the overall system of **Figure 2.42** is equivalent to an ideal band-pass filter with center frequency $-\omega_c$ and bandwidth $2\omega_0$, as illustrated in **Figure 2.48**. As the frequency ω_c of the complex exponential oscillator is varied, the center frequency of the band-pass filter varies.

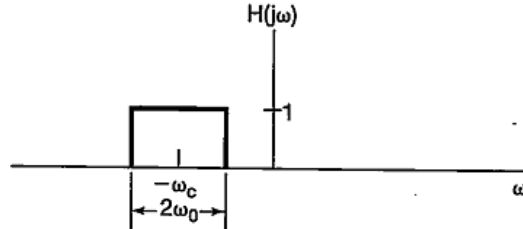


Figure 2.48. Band-pass filter equivalent of **Figure 2.11**.

In the system of **Figure 2.46** with $x(t)$ real, the signals $y(t)$, $w(t)$, and $f(t)$ are all complex. If we retain only the real part of $f(t)$, the resulting spectrum is that shown in **Figure 2.49**, and the equivalent band-pass filter passes bands of frequencies centered around ω_c and $-\omega_c$, as indicated in **Figure 2.50**.

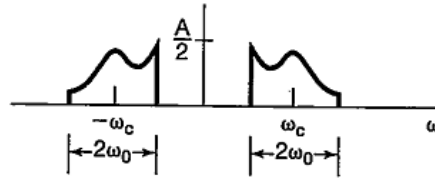


Figure 2.49. Spectrum of $\text{Re}\{f(t)\}$ associated with Figure 2.46.

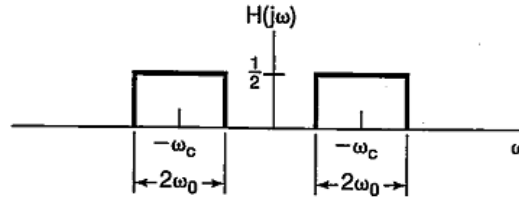


Figure 2.50. Equivalent band-pass filter for $\text{Re}\{f(t)\}$ in Figure 2.47.

One of the important applications of the frequency shifting property is amplitude modulation in communication systems. Another important application is in the implementation of frequency selective band-pass filters with tunable center frequencies that can be adjusted by the simple turn of a dial. In a frequency-selective band-pass filter built with elements such as resistors, operational amplifiers, and capacitors, the center frequency depends on a number of element values, all of which must be varied simultaneously in the correct way if the center frequency is to be adjusted directly. This is generally difficult and cumbersome in comparison with building a filter whose characteristics are fixed. An alternative to directly varying the filter characteristics is to use a fixed frequency-selective filter and shift the spectrum of the signal appropriately, using the principles of sinusoidal amplitude modulation.

2.13 More examples of Fourier Transform and usage of Fourier Transform properties

Example 2.24

Find the Fourier Transform of $x(t)$ shown in the following figure.

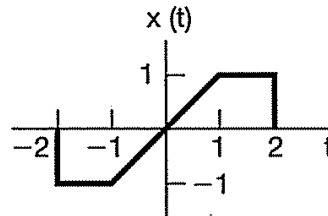


Figure 2.51 Figure for Example 2.24.

Example 2.25

The Fourier Transform of a signal $x(t)$ is shown in the following figure. What is $x(t)$?

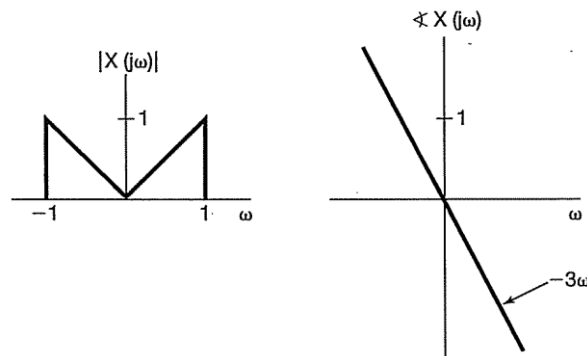


Figure 2.52 Figure for Example 2.24.

Example 2.26

Find the Fourier Transform of $x(t)$ shown in the following figure.

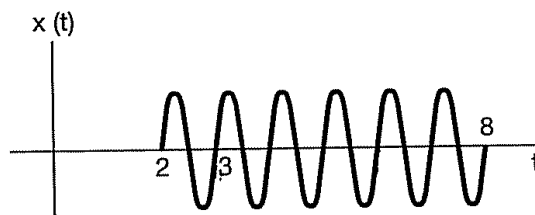


Figure 2.53. Figure for Example 2.26.

Example 2.27

Find the Fourier Transform of

$$x(t) = \frac{\sin(\pi t)}{\pi t} * \frac{\sin(2\pi(t-1))}{\pi(t-1)}.$$

Example 2.28

Find the impulse response of a system with the transfer function

$$H(\omega) = \frac{\sin^2(3\omega) \cos(\omega)}{\omega^2}.$$

Example 2.29

(a) Consider the signal

$$x_0(t) = \begin{cases} e^{-t} & 0 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}.$$

Determine the Fourier Transform of $x_0(t)$ and $x_0(-t)$.

(b) Using the results in (a), determine the Fourier Transform of the functions $x_1(t)$, $x_2(t)$ and $x_3(t)$ shown in the following figure.

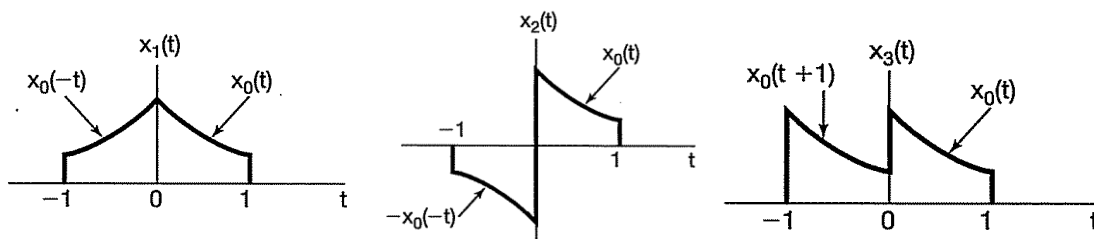
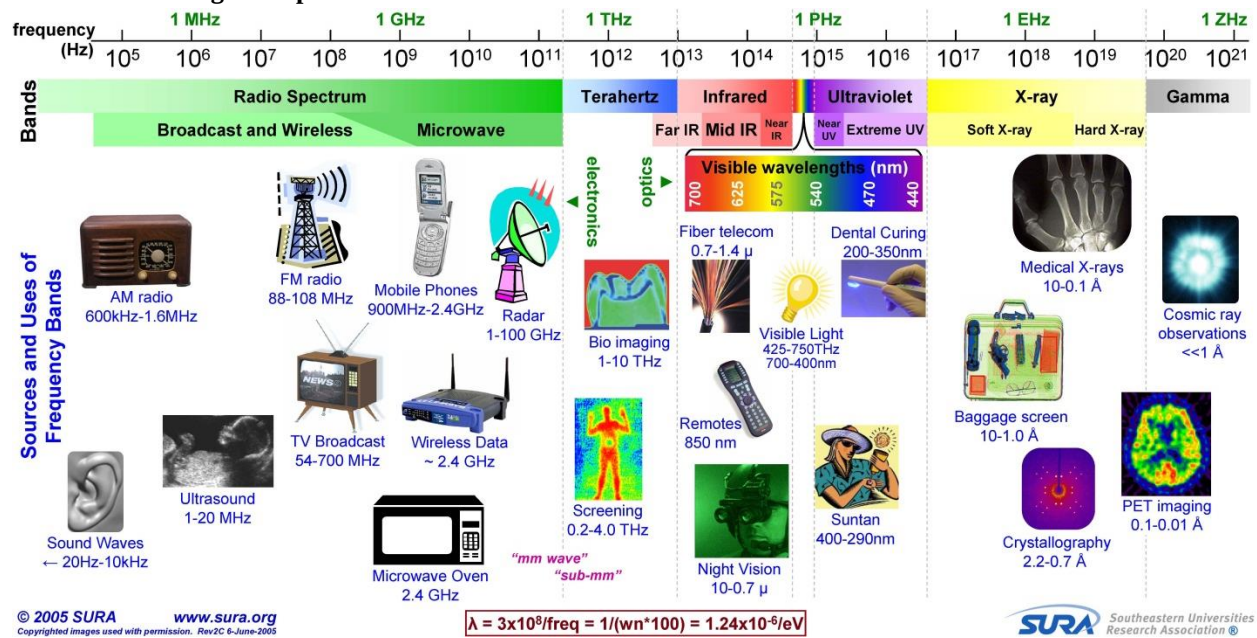


Figure 2.54. Figure for Example 2.29.

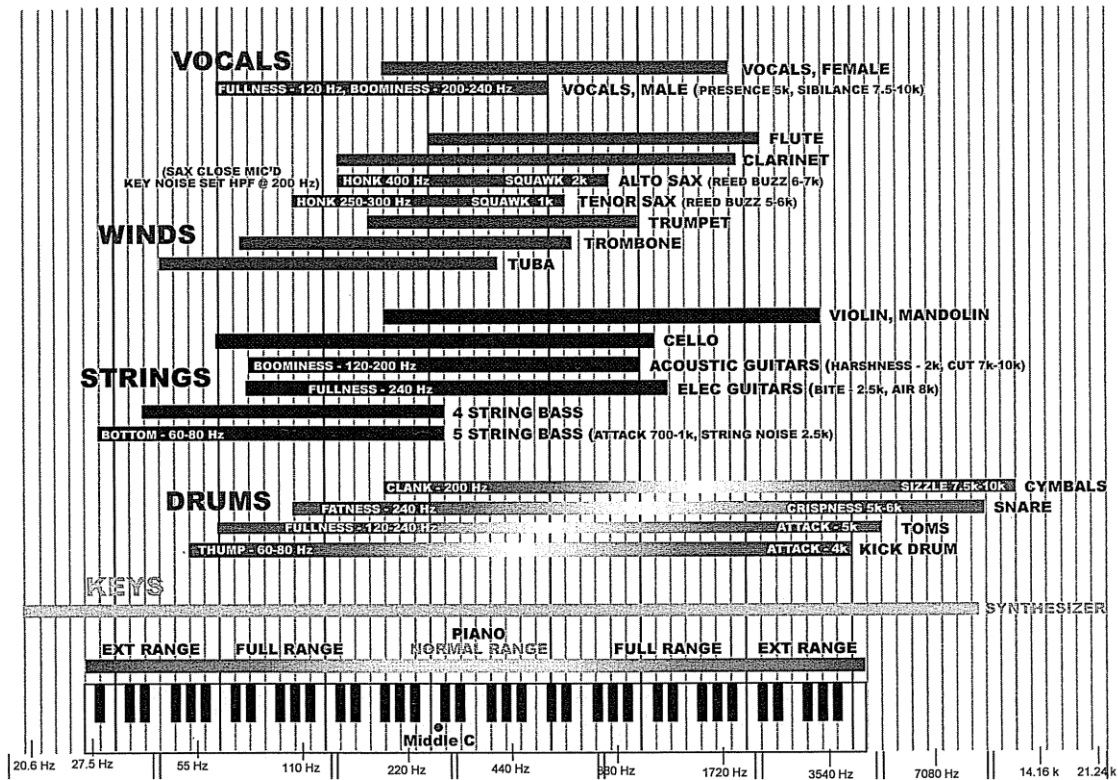
2.14 Real-life examples of signals, systems, frequency, Fourier Transform, transfer functions etc.

2.14.1. Electromagnetic spectrum



2.14.2. Music

THE FREQUENCY SPECTRUM, INSTRUMENT RANGES, AND EQ TIPS



Mixing and audio engineering

In fact, when audio engineers stand at the back and monitor the sound quality of different instruments in a concert, part of the buttons/knobs are really designing filters for different instruments to achieve certain purposes.



The following controls the amplitude of a band-pass filter centered at 12k, 2.5k and 80 Hz for different instruments.



Section	Property	Aperiodic signal	Fourier transform
		$x(t)$	$X(j\omega)$
		$y(t)$	$Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt} x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [x(t) real] $x_o(t) = \mathcal{O}\{x(t)\}$ [x(t) real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals		
	$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$		

Table 2.1 Fourier Transform properties

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0, \text{ otherwise}$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0, \text{ otherwise}$
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0, \text{ otherwise}$
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1, a_k = 0, k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \text{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T} \text{ for all } k$
$x(t) \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t), \Re\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t), \Re\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \Re\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

Table 2.2 Common Fourier Transform Pairs

Chapter 3 Amplitude Modulation

Objectives:

- To introduce amplitude modulation
- To study double sideband suppressed carrier modulation (DSB-SC) systems
- To describe coherent modulation and demodulation for DSB-SC systems
- To assess the influence of carrier phase for coherent demodulation
- To explain single sideband modulation (SSB) systems
- To study standard (conventional) amplitude modulation (AM) systems
- To describe envelope demodulation in AM system
- To introduce modulation index and illustrate its effect in AM
- To calculate sideband power, carrier power and transmission efficiency in AM
- To assess the noise performance in amplitude modulation systems
- To compare different AM systems in various aspects

Reference: Chapter 8 of *Signals and Systems* by A. V. Oppenheim and A. S. Willsky, 2nd edition, Prentice Hall 1997.

Communication Systems are a very important component of information transmission between different mediums. In order to transmit information through a communication channel, the source must first be processed by a transmitter or modulator to make the information suitable for travelling through the communication channel, then recovered by the receiver by again processing the propagated signal. The reason why information is processed is because every communication channel has a band of frequencies which it propagates for long distances with little distortion, but will attenuate or distort all other frequencies. If we look at air, the communication medium for radio signals, it propagates higher frequencies, but attenuates audible frequency range (10Hz to 20kHz). Therefore, if we want to transmit audio signals using air as the communication medium, we need to first process the signal so that it is in a higher frequency range.

Many concepts that were discussed in the previous sections of these notes play an important role for the analysis and design of communications. There are many issues to take into consideration whenever working with concepts with many applications. Although the same thing goes for communication theory, we have covered enough such that we can introduce basic issues and principles seen in the design and analysis of communication systems.

There are 3 terms worth knowing: *Modulation*, *Demodulation*, and *Multiplexing*. Modulation is the process that transforms a signal into a secondary signal, one which can be transmitted over a medium. Demodulation is the process that transforms the modulated signal back into a useable form. Multiplexing is a principle that allows multiple signals to be modulated into one signal and sent simultaneously over one channel.

3.1 Types of analog modulation

Carrier waves are usually sinusoidal, given by:

$$\text{Acos}(\omega_c t + \theta)$$

Three parameters describe this sinusoidal: amplitude A, angular frequency ω_c , and phase angle θ . These three parameters can be varied in relation to the baseband signal. **Angle modulation**, **frequency modulation**, and **phase modulation** are the modulation processes in relation to their parameters. Since frequency, and phase modulation vary the total angle of the carrier wave, the two terms are combined to form **angle modulation**.

The main aim of analog modulation (amplitude or angle) is frequency translation, to achieve frequency division multiplexing and practical antenna design, etc.

Sometimes we might tune in to a radio program, and to an AM channel. AM stands for *Amplitude Modulation*, and this chapter will go towards studying this concept, which many modulation methods are reliant upon. Amplitude modulation is the process in which a signal we want to transmit is used to modulate the amplitude of another signal. We will discuss *Sinusoidal Amplitude Modulation*, and *Frequency Division Multiplexing*. *FM (Frequency Modulation)* is the variation of frequencies instead of amplitudes for a given signal next chapter.

Voice signal frequencies range from 200 Hz to 4 kHz. In order to establish long distance microwave links, signals are required to be between 300 MHz to 300 GHz to propagate properly, whereas communication satellite links use frequencies that range from a few hundred MHz to greater than 40 GHz. Therefore, if we wanted to propagate voice signals through these channels, we would need to move the voice signals into a higher frequency. We can use sinusoidal amplitude modulation to achieve this frequency shift as we will shortly see.

Three types of AM systems will be studied in this course: ***Double sideband suppressed carrier modulation (DSB-SC), single sideband systems and conventional (standard) AM modulation systems.***

3.2 Amplitude Modulation with a Sinusoidal Carrier

A system that uses a sinusoidal carrier for amplitude modulation is depicted in **Figure 3.1**.

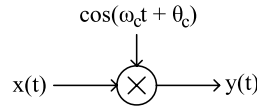


Figure 3.1. Amplitude modulation with a sinusoidal carrier.

The effect of amplitude modulation with a sinusoidal carrier can be analyzed in the following manner. For convenience we chose $\theta_c = 0$. In this case, the spectrum of the carrier signal is

$$C(\omega) = \pi [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)], \quad (3.1)$$

and thus, from the frequency translation theorem

$$Y(\omega) = \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)],$$

The spectrum of $X(\omega)$, $C(\omega)$, and $Y(\omega)$ is shown in **Figure 3.2(a)**, **(b)**, and **(c)** each respectively. If we look at $Y(\omega)$, we notice that it is the spectrum of $X(\omega)$ being centered at ω_c , and $-\omega_c$, as a result of $x(t)$ being modulated with the sinusoidal carrier $\cos(\omega_c t)$. Note that ω_c is an adjustable value depending on the sinusoidal carrier; if this value is lower than ω_M , then the two sides of $Y(\omega)$ will not be fully recoverable because they will overlap. **Figure 3.3** shows the resultant graph of $Y(\omega)$ when $\omega_c = \omega_M / 2$. The spectrum isn't fully replicated on both sides, and as such we won't be able to fully recover $x(t)$ from $y(t)$.

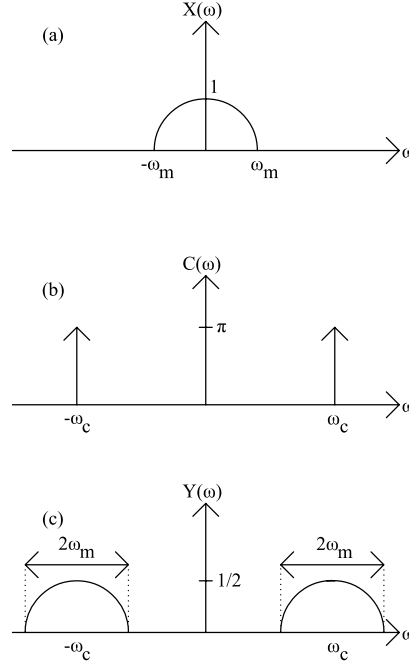


Figure 3.2. Effect in the frequency domain of amplitude modulation with a sinusoidal carrier: (a) spectrum of modulating signal $x(t)$; (b) spectrum of carrier $c(t) = \cos\omega_c t$; (c) spectrum of amplitude-modulated signal.

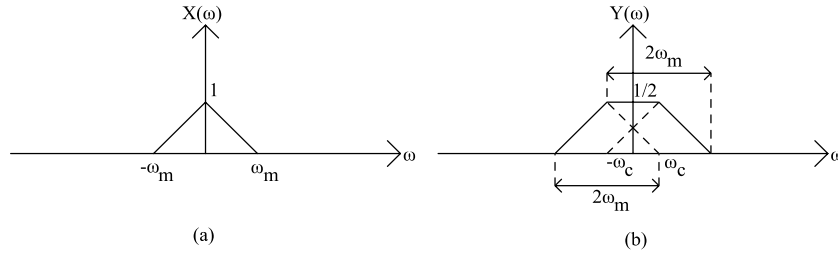


Figure 3.3. Sinusoidal amplitude modulation with carrier $\cos\omega_c t$ for which $\omega_c = \omega_m/2$: (a) spectrum of modulating signal; (b) spectrum of modulated signal.

3.3 Synchronous Demodulation For Sinusoidal AM

When the modulated signal $y(t)$ is received, it is then processed through demodulation, transforming $y(t)$ into $x(t)$ which holds information. The process of sinusoidal amplitude demodulation will be explored in this section. *Synchronous demodulation* (in-phase synchronization of transmitter, and receiver) will be described first, followed by *asynchronous demodulation*.

For *synchronous demodulation*, by assuming that $\omega_c > \omega_m$, demodulation of a signal that was modulated with a sinusoidal carrier is relatively straightforward. Specifically, consider the signal

$$y(t) = x(t) \cos \omega_c t. \quad (3.2)$$

The original signal can be recovered by modulating $y(t)$ with the same sinusoidal carrier and applying a lowpass filter to the result. To see this, consider

$$w(t) = y(t) \cos \omega_c t. \quad (3.3)$$

Figure 3.4 shows the spectra of $y(t)$ and $w(t)$, and we observe that $x(t)$ can be recovered from $w(t)$ by applying an ideal low-pass filter with a gain of 2 and a cutoff frequency that is greater than ω_m and less than $2\omega_c - \omega_m$. The frequency response of the low-pass filter is indicated by the dashed line in **Figure 3.4(c)**.

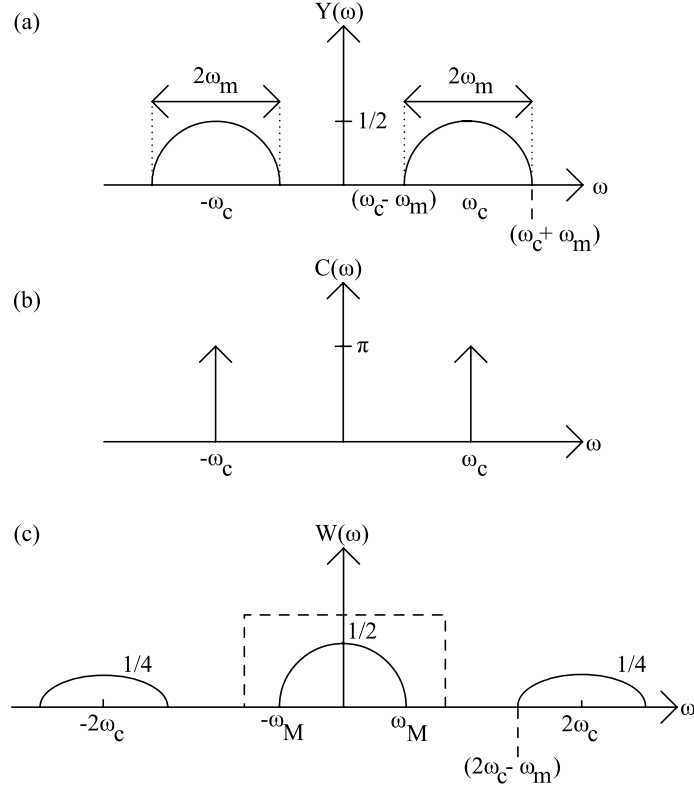


Figure 3.4. Demodulation of an amplitude-modulated signal with a sinusoidal carrier: (a) spectrum of modulated signal; (b) spectrum of carrier signal; (c) spectrum of modulated signal multiplied by the carrier. The dashed line indicates the frequency response of a low-pass filter used to extract the demodulated signal.

The basis for using eq. (3.3) and a low-pass filter to demodulate $y(t)$ can also be seen algebraically. From eqs. (3.2) and (3.3), it follows that

$$w(t) = x(t) \cos^2 \omega_c t,$$

or, using the trigonometric identity

$$\cos^2 \omega_c t = \frac{1}{2} + \frac{1}{2} \cos 2\omega_c t,$$

we can rewrite $w(t)$ as

$$w(t) = \frac{1}{2} x(t) + \frac{1}{2} x(t) \cos 2\omega_c t. \quad (3.4)$$

As we can see after the trigonometric identity is used on $w(t)$, it will have two terms, one term being a half of the original input signal, while the second one is a sinusoidal carrier with double frequency. If we look at the $W(\omega)$, the spectrum of $w(t)$, we can see that both terms are clearly shown here, with the original input signal being centered, and the sinusoidal carrier centered at double the original carrier frequencies. Applying a low-pass filter effectively eliminates the second term. This system, which uses modulation and synchronous demodulation, is known as a double sideband - suppressed carrier (DSB-SC) system.

The overall system for amplitude modulation and demodulation using a sinusoidal carrier is depicted in Figure 3.5. In these figures, we have indicated the more general case in which, for the complex sinusoidal carrier, a carrier phase θ_c is included..

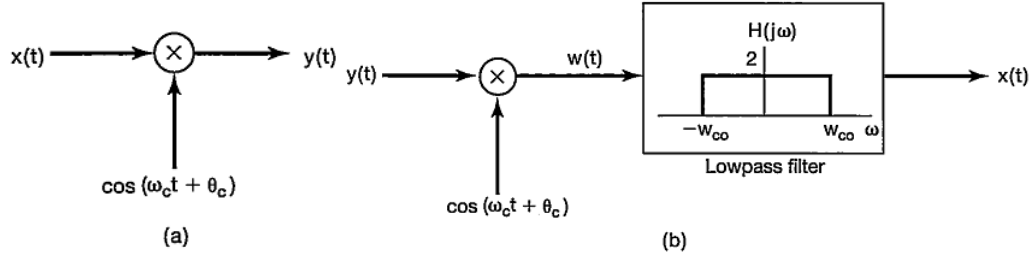


Figure 3.5. Amplitude modulation and demodulation with a sinusoidal carrier: (a) modulation system; (b) demodulation system. The lowpass filter cutoff frequency ω_{co} is greater than ω_M and less than $2\omega_c - \omega_M$.

Example 3.1 Tone modulation

The modulation signal is a pure sinusoidal, $\cos\omega_m t$.

Signal $x(t) = \cos\omega_m t$

Spectrum of $x(t)$ is given by $\pi[\delta(\omega + \omega_m) + \delta(\omega - \omega_m)]$

Modulated signal waveform $y(t) = \cos\omega_m t \cos\omega_c t = (1/2)[\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t]$

Modulated signal spectrum

$$Y(\omega) = (\pi/2)[\delta(\omega + \omega_c + \omega_m) + \delta(\omega + \omega_c - \omega_m) + \delta(\omega - \omega_c + \omega_m) + \delta(\omega - \omega_c - \omega_m)]$$

When modulated signal multiplied by a carrier $\cos\omega_c t$,

$$\begin{aligned} w(t) &= \cos\omega_m t \cos^2\omega_c t = (1/2)\cos\omega_m t (1 + \cos 2\omega_c t) = (1/2)\cos\omega_m t + (1/2)\cos\omega_m t \cos 2\omega_c t \\ &= (1/2)\cos\omega_m t + (1/4)[\cos(2\omega_c + \omega_m)t + \cos(2\omega_c - \omega_m)t] \end{aligned}$$

Spectrum of $w(t)$ is given by

$$\begin{aligned} W(\omega) &= (\pi/2)[\delta(\omega + \omega_m) + \delta(\omega - \omega_m)] + \\ &\quad (\pi/4)[\delta(\omega + 2\omega_c + \omega_m) + \delta(\omega + 2\omega_c - \omega_m) + \delta(\omega - 2\omega_c - \omega_m) + \delta(\omega - 2\omega_c + \omega_m)] \end{aligned}$$

After low-pass filtering with appropriate cutoff frequency, the spectrum of the output signal $s_o(t)$ is given by

$$S_o(\omega) = (\pi/2)[\delta(\omega + \omega_m) + \delta(\omega - \omega_m)]$$

and hence the output signal is

$$s_o(t) = (1/2)\cos\omega_m t = (1/2)x(t).$$

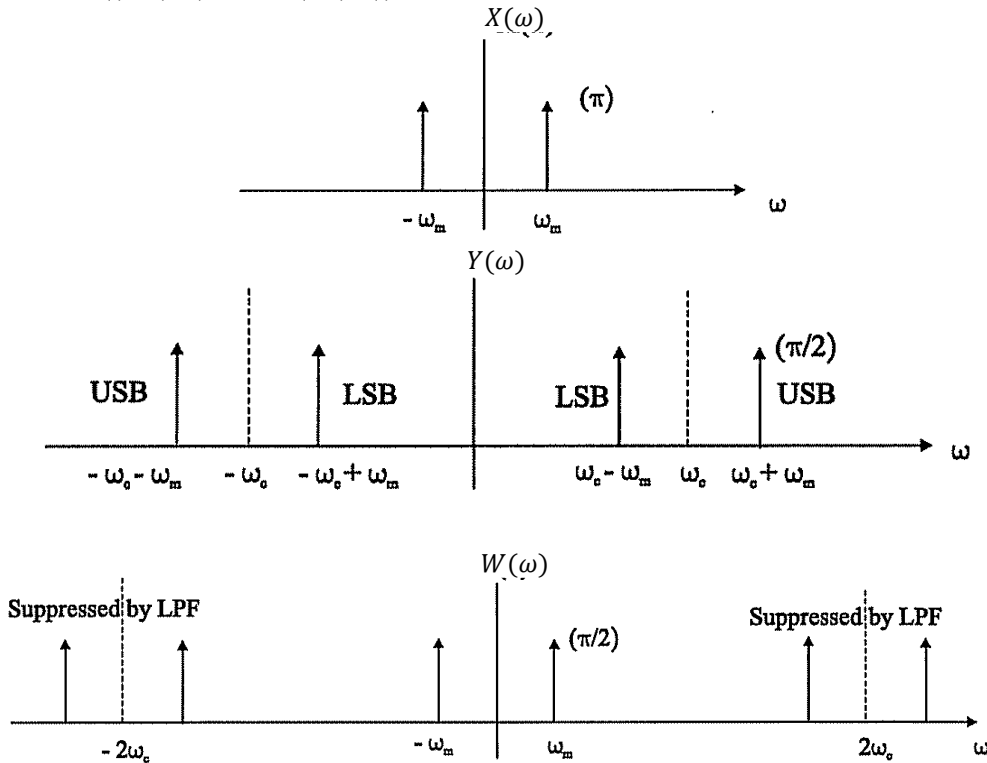


Figure 3.6. Spectra for various signals in Example 3.1.

3.4 Frequency Division Multiplexing (FDM)

With the principle of modulation and demodulation in DSB-SC systems understood, we are now able to understand how multiple signals can be simultaneously transmitted in an AM system without interference and how we can ‘tune’ to any ‘channel’ we like. This technique is known as frequency division multiplexing (FDM).

The principle of FDM is shown in the following figure.

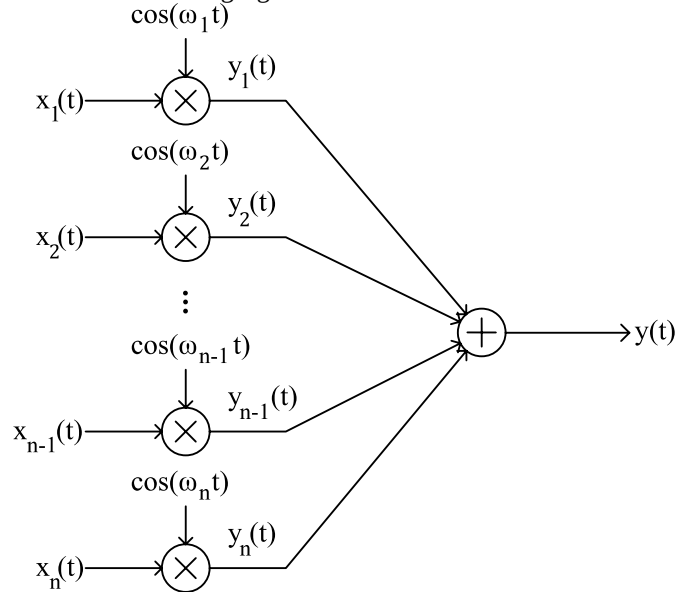
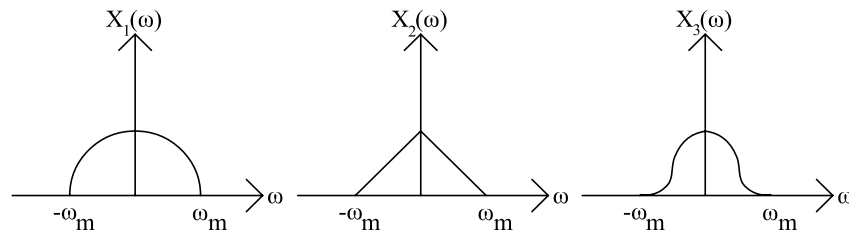


Figure 3.7. Frequency division multiplexing of three signals by modulating them with different frequencies.

In an FDM system, different signal waveforms $x_1(t)$, $x_2(t)$ and $x_3(t)$ are used to modulate carriers with distinct frequencies ω_1, ω_2 and ω_3 respectively. The modulated signals are then summed together and transmitted into free space. The transmitted signal $y(t)$ is given by

$$y(t) = x_1(t) \cos(\omega_1 t) + x_2(t) \cos(\omega_2 t) + x_3(t) \cos(\omega_3 t) .$$

Now, if the spectrum of $x_1(t)$, $x_2(t)$ and $x_3(t)$ are represented in the following figure,



then the spectrum of the transmitted signal $y(t)$ will look something like this.

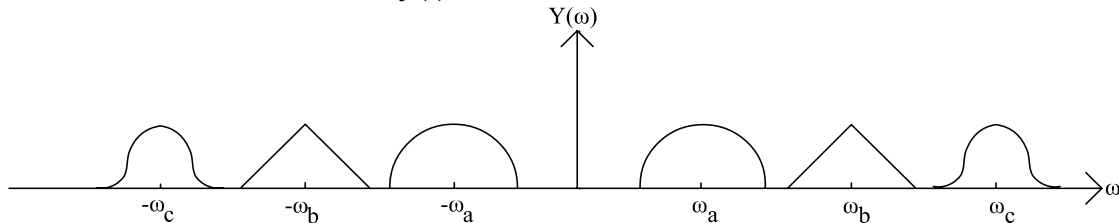


Figure 3.8. Spectrum of the multiplexed signal that is sent into a communication channel.

$Y(\omega)$, the spectrum of the transmitted signal $y(t)$, shows each original X_1 , X_2 , and X_3 in a different frequency band. To demodulate the signal we want (also called tuning to a particular channel, just like the radio), the carrier receiver at the receiver is adjusted accordingly.

How far should we separate the carrier frequencies from each other? To increase the number of channels, the carrier frequencies should be as close as possible; but if they are too close, signals might overlap each other. How can we determine the minimum carrier frequency separation?

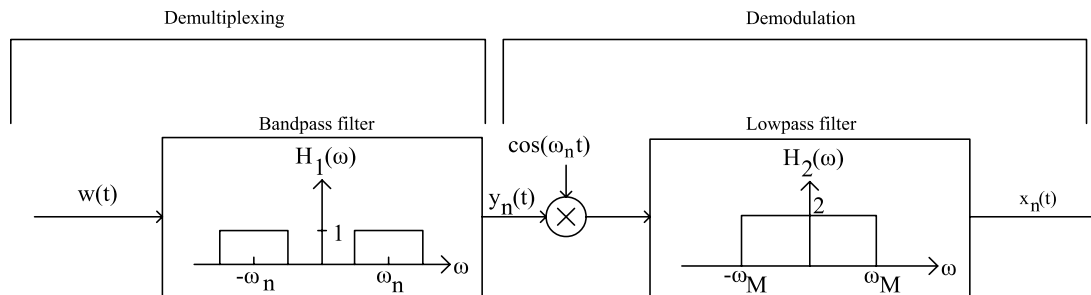


Figure 3.9. Demultiplexing and demodulation for a frequency-division multiplexed signal.

Two important applications of frequency-division multiplexing is the telephone, and signal transmission in the radio frequency band. The usage of radio frequencies is often controlled by a country or region. In the United States, the transmission of signals from the 10 kHz to 275 GHz is regulated by the FCC (The Federal Communications Commission), and different frequency ranges are allocated for different uses, shown in Figure 8.18. The figure shows that the AM is used near the 1 MHz frequency range. AM radio stations are given a frequency within this AM band, and many stations can broadcast by using frequency division multiplexing. The receiver, a radio of some sort, uses demodulation, and demultiplexing to tune into an individual radio station. The tuning dial would control the centre frequency of the bandpass filter, and the frequency of the demodulating oscillator. Public broadcasting, in actuality, uses asynchronous (not synchronous) modulation, and demodulation to reduce costs of receivers (radios). Frequency-division multiplexed signals require sharp cutoff bandpass filters with variable center frequency in order for demultiplexing, and demodulation. Variable frequency-selective filters are difficult to implement, so a fixed filter is implemented in its place, and an intermediate stage of modulation and filtering (also known as the *intermediate-frequency* (IF) stage in radio receivers) is used. Modulation is replaced the need for a variable filter because it can shift the spectrum of the signal to the fixed bandpass filter. This is incorporated into typical AM radio receivers.

As seen in **Figure 3.7**, the frequency-division multiplexing system replicates the original signal in both the negative and positive frequencies, so each modulated signal occupied twice the bandwidth of the original. This is inefficient, and in the next section we will consider an alternative form of sinusoidal amplitude modulation, which saved bandwidth at the expense of a more complex system.

Frequency range	Designation	Typical uses	Propagation method	Channel features
30-300 Hz	ELF (extremely low frequency)	Macrowave, submarine communication	Megametric waves	Penetration of conducting earth and seawater
0.3-3 kHz	VF(voice frequency)	Data terminals, telephony	Copper wire	
3-30 kHz	VLF (very low	Navigation, telephone, telegraph,	Surface ducting	Low attenuation, little fading, extremely stable

	frequency)	frequency and timing standards	(ground wave)	phase and frequency, large antennas
30-300 kHz	LF (low frequency)	Industrial (power line) communication, aeronautical and maritime long-range navigation, radio beacons	Mostly surface ducting	Slight fading, high atmospheric pulse
0.3-3 MHz	MF (medium frequency)	Mobile, AM broadcasting, amateur, public safety	Ducting and ionospheric reflection (sky wave)	Increased fading, but reliable
3-30 MHz	HF (high frequency)	Military communication, aeronautical mobile, international fixed, amateur and citizen's band, industrial	Ionospheric reflecting sky wave, 50-400 km layer altitudes	Intermittent and frequency-selective fading, multipath
30-300 MHz	VHF (very high frequency)	FM and TV broadcast, land transportation (taxis, buses, railroad)	Sky wave (ionospheric and tropospheric scatter)	Fading, scattering, and multi-path
0.3-3 GHz	UHF (ultra high frequency)	UHF TV, space telemetry, radar, military	Transhorizon tropospheric scatter and line-of-sight relaying	
3-30GHz	SHF (super high frequency)	Satellite and space communication, common carrier (CC), microwave	Line-of-sight ionosphere penetration	Ionospheric penetration, extraterrestrial noise, high directly
30-300GHz	EHF (extremely high frequency)	Experimental, government, radio astronomy	Line of sight	Water vapor and oxygen absorption
10^3 - 10^7 GHz	Infrared, visible light, ultraviolet	Optical communications	Line of sight	

Table 3.1 Allocation of frequencies in the RF spectrum.

Example 3.2 Frequency Division Multiplexing

Multiple signals $m_1(t)$, $m_2(t)$, $m_3(t)$... each with bandwidth 5000 rad/s are to be transmitted simultaneously over a channel by FDM. If you are allowed to use the radio spectrum from 15k rad/s – 75k rad/s, how many of these signals channels can you send at the same time?

3.5 Influence of carrier phase for coherent demodulation

In DSB-SC system, the demodulation process consists of multiplying the incoming signal by a local carrier of the same frequency and phase with the carrier at the modulator. The product is then passed through a low pass filter in order to recover the signal. However, this task is difficult to achieve or quite expensive.

Suppose, however, that the modulator and demodulator are not synchronized in phase. For the sinusoidal carrier, let θ_c and ϕ_c denote the phases of the modulating and demodulating carriers, respectively, as indicated in **Figure 3.10**. Sinusoidal amplitude modulation and demodulation system for which the carrier signals and the modulator and demodulator are not synchronized: (a) modulator; (b) demodulator.. The input to the low-pass filter is now

$$w(t) = x(t) \cos(\omega_c t + \theta_c) \cos(\omega_c t + \phi_c), \quad (3.5)$$

or, using the trigonometric identity

$$\cos(\omega_c t + \theta_c) \cos(\omega_c t + \phi_c) = \frac{1}{2} \cos(\theta_c - \phi_c) + \frac{1}{2} \cos(2\omega_c t + \theta_c + \phi_c), \quad (3.6)$$

we have

$$w(t) = \frac{1}{2} \cos(\theta_c - \phi_c) x(t) + \frac{1}{2} x(t) \cos(2\omega_c t + \theta_c + \phi_c), \quad (3.7)$$

and the output of the lowpass filter is then $x(t)$ multiplied by the amplitude factor $\cos(\theta_c - \phi_c)$. If the oscillators in the modulator and demodulator are in phase, $\theta_c = \phi_c$, and the output of the lowpass filter is $x(t)$. On the other hand, if these oscillators have a phase difference of $\pi/2$, the output will be zero. In general, for a maximum output signal, the oscillators should be in phase. Of even more importance, the phase relation between the two oscillators must be maintained over time, so that the amplitude factor $\cos(\theta_c - \phi_c)$ does not vary. This requires careful synchronization between the modulator and the demodulator, which is often difficult, particularly when they are geographically separated, as is typical in a communication system. In general, the carrier frequency of the transmitter and receiver is also different so that the low-pass filter output is given by

$$w(t) = \frac{1}{2} \cos(\Delta\omega t + \theta_c - \phi_c) x(t)$$

where $\Delta\omega$ is the different in carrier frequency between the transmitter and receiver. The corresponding effects of, and the need for, synchronization not only between the phase of the modulator and demodulator, but between the frequencies of the carrier signals used in both, are explored in more detail in the reference books. In general, $\Delta\omega$ and $\theta_c - \phi_c$ varies with time and additional synchronizing circuits that ‘track’ the changes in $\Delta\omega$, $\theta_c - \phi_c$ are used in practical systems. These circuits are called phase lock loops.

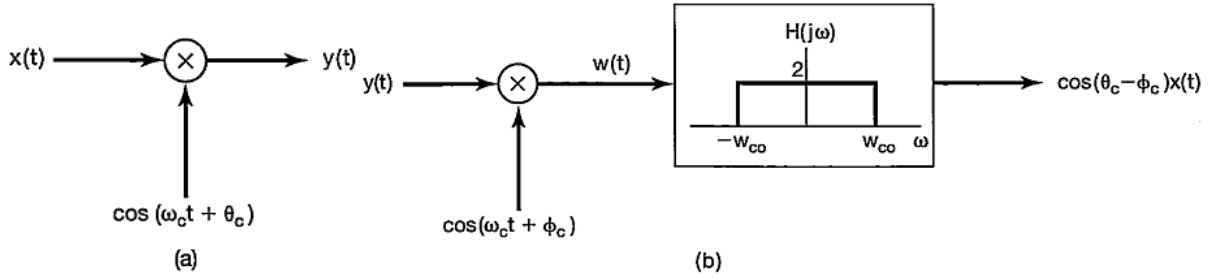


Figure 3.10. Sinusoidal amplitude modulation and demodulation system for which the carrier signals and the modulator and demodulator are not synchronized: (a) modulator; (b) demodulator.